Population Growth and CO_2 Emissions: A Worldwide Analysis

Data-analysis report prepared in R (tidyverse)

November 6, 2026

Abstract

This study examines whether the size of a country's population is related to its carbon dioxide (CO_2) emissions, using the Kaggle *World Population Dataset* (234 countries and territories, 17 variables) and the *CO2 Emission Around the World* dataset (per-capita emissions, 1990–2019). The data were managed and analysed entirely in R with the tidyverse. After cleaning—standardising column names, removing duplicates, imputing missing values with the median, and winsorising rate variables by the inter-quartile range—an exploratory analysis described the structure and quality of the population data. A new variable, each country's projected 2030 population, was created with the exponential growth model $N_t = Pe^{rt}$; the world total is projected to rise from 7.97 billion in 2022 to 8.57 billion in 2030 (+7.5%). The ten most populous countries and their 1990–2022 population and 1990–2019 emission trends were visualised with ggplot2. Correlation and linear regression were then used on 172 countries matched across both datasets. The association between 2022 population and 2019 per-capita emissions was found to be negligible (Pearson $r = 0.015$; regression slope -0.171, $p = 0.67$, $R^2 \approx 0.001$): population size does not predict per-capita emissions. Summing 2019 per-capita emissions by continent placed Asia first and South America last. The main lesson is that emission intensity is driven by economic and energy structure, not by headcount, which has direct implications for how climate responsibility is interpreted.

1 Introduction

Population growth is often cited as a driver of rising greenhouse-gas emissions. Yet emissions depend not only on *how many* people live in a country but on *how* they produce and consume energy. This report tests the population –emissions link empirically. The objectives follow the assignment brief: (i) manage and clean the population data; (ii) perform exploratory data analysis (EDA); (iii) engineer a projected-population variable; (iv) visualise population and emission trends for the largest countries; (v) quantify correlations, including a merged population–emissions analysis; and (vi) compare emissions across continents.

2 Methods

2.1 Datasets

The **World Population Dataset** contains 234 rows and 17 columns: rank, country code (CCA3), country/territory, capital, continent, population snapshots for 1970, 1980, 1990, 2000, 2010, 2015, 2020 and 2022, land area (km^2), density, growth rate and world-population percentage. The **CO2 Emission Around the World** dataset gives annual CO_2 emissions in metric tons per capita for each country from 1990 to 2019 (the file shipped a duplicated 2019 column, which was de-duplicated during loading).

2.2 Tools and procedure

All work was done in R 4.6 using the tidyverse (`readr`, `dplyr`, `tidyr`, `ggplot2`) with `janitor`, `scales` and `corrplot`, and knitted to HTML with `rmarkdown`/`knitr`. Cleaning used `clean_names()` for consistent naming, `distinct()` for duplicate removal, median imputation for missing numeric values, and IQR winsorising for the rate variables (density, growth rate, world-population percentage) used in the correlation study; genuine large-country values in the population columns were retained. Box plots were used to inspect outliers. The 2030 projection used the exponential model

$$N_t = P\,e^{rt}, \qquad P = 2022 \text{ population},\ t = 8,\ r = (\text{growth-rate factor}) - 1.$$

The two datasets were merged on country name (after normalising well-known name differences such as "Russian Federation" → "Russia"), retaining 2022 population and 2019 emissions for 172 matched countries.

3 Results

3.1 EDA and projection (Tasks 4.2–4.3)

The population data were complete after cleaning, with no duplicate rows. Population columns are strongly right-skewed—China and India are genuine extremes rather than errors. Applying the growth model, the world population is projected to grow from 7 973 413 042 (2022) to 8 569 157 744 (2030). Growth is highly uneven: Nigeria is projected to grow by about 21%, while China, whose growth rate is effectively zero, stays flat and is overtaken in momentum by India (Table 1).

Table 1: Ten most populous countries: 2022 population and projected 2030 population.

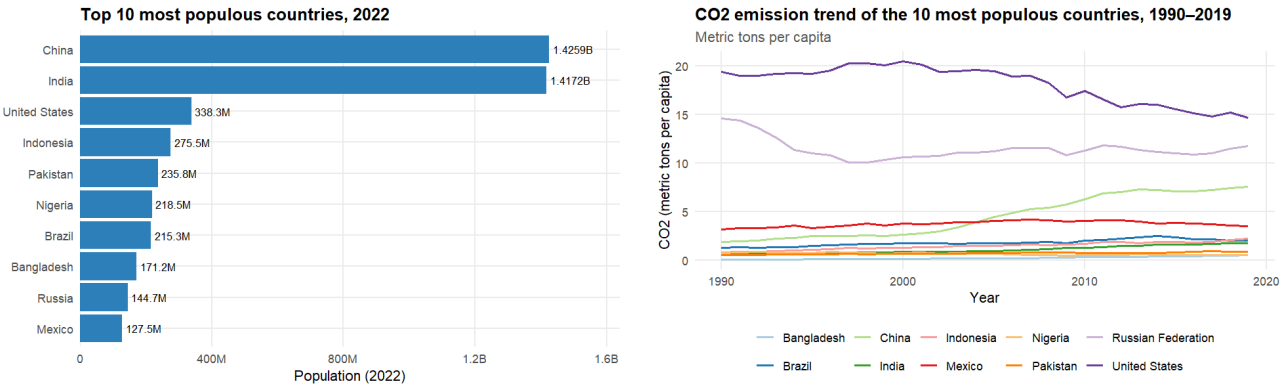
Country	Continent	2022 population	Projected 2030
China	Asia	1 425 887 337	1 425 887 337
India	Asia	1 417 173 173	1 496 402 904
United States	North America	338 289 857	348 731 782
Indonesia	Asia	275 501 339	289 974 355
Pakistan	Asia	235 824 862	274 757 645
Nigeria	Africa	218 541 212	265 011 868
Brazil	South America	215 313 498	223 384 633
Bangladesh	Asia	171 186 372	186 634 630
Russia	Europe	144 713 314	141 621 023
Mexico	North America	127 504 125	134 095 029

3.2 Trends for the largest countries (Task 4.4)

Bar and line charts (see the accompanying `analysis.html`) show that the populations of the ten largest countries rose steadily from 1990 to 2022, led by China and India. Their per-capita emission trends (1990–2019) diverge sharply: the United States has high but gradually declining per-capita emissions, China's rose steeply before levelling, while India, Nigeria and Bangladesh remain low per capita despite very large populations.

3.3 Trends for the largest countries (Task 4.4)

Figure 1 shows that the populations of the ten largest countries rose steadily from 1990 to 2022, led by China and India. ...



(a) Population, 1990–2022

(b) CO_2 per capita, 1990–2019

Figure 1: Population and per-capita emission trends for the ten most populous countries.

3.4 Correlation and regression (Task 4.5)

Among the population variables, pairwise correlations between area, density, growth rate and world-population percentage are weak, the clearest being a positive link between land area and world-population share. For the merged data, the relationship between 2022 population and 2019 per-capita emissions is *negligible*: Pearson $r = 0.015$, Spearman $\rho = -0.088$, and a linear regression of per-capita emissions on $\log_{10}$(population) gives a slope of -0.171 ($p = 0.67$) with $R^2 \approx 0.001$. Population size therefore explains virtually none of the variation in per-capita emissions.

3.5 Continental comparison (Task 4.6)

Table 2 sums 2019 per-capita emissions by continent. **(1)** The highest-emitting continent is **Asia** (261.42 metric tons per capita summed). **(2)** The third continent is **North America**, totalling 79.57 metric tons per capita. **(3)** The lowest is **South America**, at 29.42 metric tons per capita. Because this is a *sum* of per-capita values, the totals are influenced by how many countries each continent contributes; on a per-country mean basis Asia (6.70) and Europe (5.88) remain highest and Africa lowest (1.23).

Table 2: Total and mean 2019 CO_2 (metric tons per capita) by continent, matched countries.

Continent	Total	Mean	Countries
Asia	261.42	6.70	39
Europe	240.97	5.88	41
North America	79.57	4.19	19
Africa	60.50	1.23	49
Oceania	53.20	4.09	13
South America	29.42	2.67	11

4 Discussion

The headline result is that a country's population size is essentially unrelated to its per-capita CO_2 emissions. This is consistent with climate science: per-capita emissions reflect income levels, energy mix and industrial structure rather than headcount. Highly populous but lower-income countries (India, Nigeria, Bangladesh) have low per-capita emissions, whereas smaller, energy-intensive economies emit far more per person. The continental comparison must be read with care, because summing a per-capita rate rewards continents with many countries; the per-country means give a fairer ranking and still show Asia and Europe ahead of Africa.

5 Lessons learned and recommendations

The project reinforced several practical points. First, *data quality work dominates*: the duplicated 2019 column silently emptied the merge until it was handled, showing why loading should always be validated. Second, *units matter*—summing a per-capita metric answers the literal question but can mislead; totals and means should be reported together. Third, *robust code* (auto-detecting files, matching year columns by pattern, normalising country names) makes an analysis reproducible. Recommendations: (i) analyse *total* (not only per-capita) emissions to assess absolute climate impact; (ii) bring in GDP and energy-mix variables, which should explain emissions far better than population; and (iii) reconcile country naming with ISO codes before merging to avoid losing records.

6 Conclusion

Using the tidyverse, this study cleaned and projected world population data, visualised trends for the largest countries, and tested the population–emissions link. Population is projected to reach 8.57 billion by 2030, but its size shows no meaningful correlation with per-capita emissions ($R^2 \approx 0.001$). Emission intensity is a function of how economies are powered, not of how many people they hold—an important distinction for interpreting national climate responsibility.